Multi-Diagnostic Ensemble for Physics-Informed Neural Networks

Ridham Patel[1]

Computer Science & Engineering, Indian Institute of Technology Gandhinagar, India
ridham.patel@iitgn.ac.in

Abstract. Physics-informed machine learning fails where there is multiscale structure, stiffness, and localized sharp features. Unidirectional analysis to minimize these errors is often not enough due to multiscale residual components. Therefore, we introduce an ensemble method that performs various diagnostics and solves the problem based on the collective opinion of its components. This Method is tested and proven over several PDEs, including the Allen-Cahn Equations, which construct all the required challenges that we plan to mitigate.

Keywords: Ensemble Learning · Multi-Scale Features · Physics-Informed Neural Networks · Stiff Systems

1 Introduction

Physics-informed machine Learning has recently emerged as a prominent solution to traditional numerical methods that required heavy computing. Thereafter, they have been used as a prominent method for solving forward and inverse problems governed by partial differential equations. It is usually achieved by embedding the residuals of governing laws into the loss function of neural networks. Offering mesh free solution is an added advantage of the Physics Informed Machine Learning. Despite their success, they have struggled in performing accurately in certain areas, leading to systematic errors. Spectral bias in neural networks has been a universal bottleneck, but it becomes important to solve when it comes to scientific computing or machine learning [15]. These systematic errors also include loss-term imbalance that suppresses constraints, under-resolved sharp interfaces due to stiffness, and diagnostic blind spots when relying on a single residual norm or metric.

Among various such examples, the Allen-Cahn equation is a prime example of a PDE that highlights these flaws in training dynamics. This equation exhibits bistable wells with narrow diffuse interfaces whose width scales with $\sqrt{\epsilon}$, making accurate resolution sensitive to small parameter changes and creating multiscale residual content across training. There have been several works that try to fix this by analyzing a single aspect of this residual, like Liu et al., 2025, who performed spectral analysis [9], while Chu et al., 2024, preserved structure [4]. While these methods show promising results, they are still not enough for

maximal elimination of the errors. Therefore, in this paper, we introduce an architecture that uses an ensemble of various novel analyzers to minimize the loss.

1.1 Contributions

The contributions of this work can be summarized as follows:

1. Proposes an architecture that aggregates opinions from topological persistence proxies, manifold complexity estimators (intrinsic dimensionality, curvature, clustering), geometric-flow diagnostics (gradient magnitude, Laplacian divergence, vorticity proxy, defect counts), and FFT-based spectral analyzers.
2. Introduces an EnsembleAnalyzerLoss term that augments the standard loss components, combining PDE loss with boundary and initial condition losses.
3. Employs adaptively weighted loss components, where different parts of the objective (PDE, BC, IC) receive time-varying weights to maintain balance during training.
4. Designs a SpectralAnalyzer that computes log-spaced band energies and a high-frequency ratio, which in turn drives an AdaptiveLossWeighter that increases PDE emphasis when high-frequency residual content surges.

2 Related work

Classical numerical solvers for Allen-Cahn. The Allen-Cahn equation is classically treated with finite difference, finite element, and spectral methods that address stiffness via implicit or semi-implicit time integration and energy-stable discretizations, often leveraging interface-width scaling with $\sqrt{\epsilon}$ for mesh and timestep selection [19, 2]. Structure-preserving designs such as convex splitting and stabilized IMEX schemes provide unconditional energy stability, enabling robust long-time simulations for phase-field dynamics [19, 18].

PINNs and SciML foundations. Physics-Informed Neural Networks (PINNs) unify PDE residuals, boundary & initial conditions, and optional data into a single objective computed with automatic differentiation, supporting mesh-free training and flexible geometries [16]. Subsequent surveys and libraries organize practice and highlight common training pathologies like spectral bias, stiffness, while providing reproducible toolchains [8, 10].

Single-axis remedies: spectral, weighting, and structure. Spectral bias remedies enrich representations with Fourier features or frequency-aware designs to better capture high-frequency content without changing task couplings [20, 17]. Loss-imbalance remedies learn adaptive weights among PDE/BC/IC terms to equilibrate gradients and stabilize optimization [21, 11, 13]. Structure-preserving approaches encode invariants like Hamiltonian,symplectic or energy dissipation to improve physical fidelity along that axis [22].

Allen–Cahn with PINNs. Allen–Cahn serves as a stringent benchmark for learning solvers due to sharp diffuse interfaces, metastability, and defect dynamics; single-axis methods often improve one failure mode but leave others like spectral vs. geometric/topological partially unaddressed [17, 11, 22]. This motivates designs that combine frequency-aware adaptation with complementary diagnostics and principled weighting.

3 Problem setup

Governing equation and domain. We study the Allen-Cahn equation on a rectangular space-time domain $\Omega \subset R^2$ with time horizon $[0, T]$: $u_t = \varepsilon \Delta u - \alpha (u^3 - u)$, $(x, y, t) \in \Omega \times [0, T]$, With prescribed boundary and initial conditions. The parameters $\varepsilon > 0$ and $\alpha > 0$ control interface thickness and double-well strength and are treated as known hyperparameters throughout. (It must be noted that our work is not specific to Allen-Cahn Equation. For the sake of detailed analysis, we have included the Allen-Cahn equation in the main text. Although we have also studied various other PDEs, as shown in Appendix C.)

Neural parameterization. A fully-connected network $f_\theta : R^3 \to R$ with tanh activations maps inputs (x, y, t) to the scalar field u, i.e., $u = f_\theta(x, y, t)$. Spatial and temporal derivatives used in the PDE residual are computed via automatic differentiation with nested gradient calls to obtain u_t, u_x, u_y, u_{xx}, and u_{yy}. The Laplacian is $\Delta u = u_{xx} + u_{yy}$.

Physics residual. For a batch of collocation inputs $X_{\mathcal{C}} = \{(x_i, y_i, t_i)\}$, the Allen-Cahn residual reads

$$r(x, y, t) = u_t - \varepsilon (u_{xx} + u_{yy}) - \alpha (u^3 - u). \tag{1}$$

All derivatives are obtained by autodiff while retaining the graph where needed for second derivatives.

Data sets. Training uses four disjoint sets: (i) Collocation points $\mathcal{C}$, (ii) boundary points $\mathcal{B}$ with labels u_{bc}, (iii) initial points $\mathcal{I}$ with labels u_{ic} at $t = 0$, (iv) optional supervised samples $\mathcal{S}$ with labels v.

If an explicit $\mathcal{S}$ is not provided, a supervised pathway automatically assembles labelled pairs from available BC/IC (and optionally labelled collocation) to form a data-driven consistency term. Device/dtype/shape coercions, as well as sample-count diagnostics, are handled robustly in our experimentation.

4 Method

Our approach augments a standard physics-informed neural network (PINN) for a given PDE with two principled mechanisms: (i) adaptive spectral weighting driven by residual frequency content, and (ii) a learnable ensemble of scientific

analyzers producing a differentiable guidance signal that shapes optimization. The full objective combines PDE, boundary, initial, and supervised terms with an auxiliary ensemble coupling. All components are modular and dependency-light, with fallbacks to ensure robustness.

We employ a fully-connected multilayer perceptron $f_\theta : R^3 \to R$ with tanh activations. Inputs are (x, y, t) and the output is the scalar field u. A typical configuration uses L hidden layers of width W; weights/biases are initialized with standard Xavier/Glorot schemes. The forward map is $z_0 = [x, y, t], \quad z_{\ell+1} = \tanh(W_\ell z_\ell + b_\ell), \quad u = W_L z_L + b_L$, with derivatives computed by automatic differentiation. First and second spatial derivatives produce $\Delta u = u_{xx} + u_{yy}$, while the time derivative yields u_t, enabling residual computation. Domain bounds are configurable and used by analyzers for gridding and slicing.

4.1 PDE residual and base losses

At collocation points $\mathcal{C}$, specifically for the Allen–Cahn, residual is

$$r(x, y, t) \;=\; u_t \;-\; \varepsilon\,(u_{xx} + u_{yy}) \;-\; \alpha\,(u^3 - u). \tag{2}$$

We define

$$\mathcal{L}_{\text{pde}} = E_{\mathcal{C}}[r^2], \quad \mathcal{L}_{\text{bc}} = E_{\mathcal{B}}[(u - u_{\text{bc}})^2], \quad \mathcal{L}_{\text{ic}} = E_{\mathcal{I}}[(u - u_{\text{ic}})^2]. \tag{3}$$

An automatic supervised pathway forms $\mathcal{L}_{\text{sup}} = E_{\mathcal{S}}[(u - v)^2]$ using provided supervised samples; when absent, labeled pairs are assembled from BC/IC (and optional labeled collocation). Shapes, device, and dtype are coerced safely, and sample counts are logged.

4.2 Adaptive loss weighting

An AdaptiveLossWeighter updates the weights $w_{\text{pde}}, w_{\text{bc}}, w_{\text{ic}}$ using ρ (high frequency share of the residual via band-energy aggregation on re-sampled slices) and current loss magnitudes. When ρ exceeds a threshold (indicating elevated high-frequency residual content), w_{pde} increases, while w_{bc} and w_{ic} are maintained via time-scaled floors to avoid vanishing constraints. The weights are normalized to a fixed sum each update. A supervised weight w_{sup} activates when supervised data are present. All weights and ρ are logged per epoch to enable ablations and reproducibility.

4.3 Ensemble of analyzers

We introduce a suite of analyzers, each outputting a scalar boost b_k that reflects diagnostic evidence of underfit:

Topology-inspired residual complexity proxies. We estimate the topological complexity of the residual field to detect defects or cycles without requiring heavyweight persistent homology dependencies. Importantly, this module provides a topology-inspired proxy of residual complexity rather than computing formal persistent homology. Let $\mathcal{R} = \{(x_i, y_i, t_i, r_i)\}_{i=1}^{N}$ denote residual samples and let $\mathbf{z}_i = [x_i, y_i, t_i, r_i]$. We calculate statistical proxies, TopoVar $=$ $\mathrm{Var}(r_i)$, AbsMean $= \frac{1}{N}\sum_{i=1}^{N}|r_i|.$, that capture overall fluctuation magnitude and bias-free activity. Additionally, we calculate a PH-inspired proxy where we form a point cloud $\mathcal{P} = \{[x_i, y_i, r_i]\}$ and compute a Rips complex up to dimension 1 [5, 6].

We define CycleStrength $\approx \frac{1}{M}\sum_{m=1}^{M}\max\left(0,\ \overline{r}_{\partial\mathcal{N}_m} - \overline{r}_{\mathcal{N}_m}\right)$, where $\mathcal{N}_m$ is a small neighborhood and $\partial\mathcal{N}_m$ its boundary ring; this approximates persistent loop contrast.

On the top of that, we calculate connectivity proxy. On a gridded residual $R \in R^{N_x \times N_y}$ at fixed t, threshold R at τ to produce a binary mask $B = 1\{|R| > \tau\}$ and count connected components #CC, normalized by grid size. Collectively, we can write the composite topological score as,

$$S_{\text{topo}} \;=\; a_1\,\text{TopoVar} \;+\; a_2\,\text{AbsMean} \;+\; a_3\,\text{CycleStrength} \;+\; a_4\,\frac{\#\text{CC}}{N_x N_y}, \quad (4)$$

Positive coefficients a_j is chosen for scale balance. Finally, the boost is expressed as

$$b_{\text{topo}} \;=\; 1 \;+\; \min\left\{2,\ c_{\text{topo}}\,\frac{S_{\text{topo}}}{S_{\text{topo}} + \kappa}\right\}. \quad (5)$$

Manifold complexity We estimate residual manifold complexity from joint embeddings of coordinates and residuals to capture intrinsic dimension, curvature, and clustering. After forming features $\mathbf{f}_i = [x_i, y_i, t_i, r_i]$, we define Intrinsic dimension proxy via SVD as:

$$\text{ID} \;=\; \sum_{j=1}^{d}\frac{\sigma_j^2}{\sum_{\ell=1}^{d}\sigma_\ell^2}, \quad \text{for the smallest } d \text{ such that } \sum_{j=1}^{d}\sigma_j^2 \geq \eta\sum_{\ell}\sigma_\ell^2, \quad (6)$$

with $\eta \in [0.90, 0.99]$. Larger d implies greater intrinsic complexity. Curvature proxy is calculated by embedding $\mathbf{f}_i$ to low dimension $\tilde{\mathbf{f}}_i$ via UMAP [12] and compare pairwise distances as:

$$\mathcal{C} \;=\; \frac{1}{N^2}\sum_{i,j}\left|\,\|\mathbf{f}_i - \mathbf{f}_j\|_2 \;-\; \|\tilde{\mathbf{f}}_i - \tilde{\mathbf{f}}_j\|_2\,\right|, \quad (7)$$

higher values indicate stronger nonlinearity/curvature. Additionally, cluster complexity for a cluster $\mathbf{f}_i$ (DBSCAN) is calculated to get the number of clusters K_c and outlier fraction p_{out} [7]. Define

$$\text{Clust} \;=\; \frac{K_c}{1 + \log N} \;+\; p_{\text{out}}. \quad (8)$$

Composite manifold score is defined as, $S_{\text{mani}} = b_1\,\text{ID} + b_2\,\mathcal{C} + b_3\,\text{Clust}$, with nonnegative b_j. Finally, the boost can be expressed as:

$$b_{\text{mani}} = 1 + \min\Big\{2,\ c_{\text{mani}}\,\frac{S_{\text{mani}}}{S_{\text{mani}}+\kappa}\Big\}.$$

Geometric-flow diagnostics We capture differential-geometric features of the residual field on gridded space–time slices to detect sharp fronts, sources, sinks, and vortex-like signatures. Interpolate residual samples to a regular grid (x_p, t_q) at fixed y or to (x_p, y_q) at fixed t using linear/bilinear interpolation. Denote the gridded residual by R. Compute discrete gradients and Laplacians:

$$\nabla R = (R_x, R_y \text{ or } R_t), \quad \|\nabla R\| = \sqrt{R_x^2 + R_y^2 \text{ or } R_t^2},$$

$$\Delta R = R_{xx} + R_{yy} \text{ or } R_{xx} + R_{tt}.$$

This Analyzer calculates certain metrics, like Gradient magnitude, which can be expressed as $G = \frac{1}{|\mathcal{G}|}\sum_{(p,q)\in\mathcal{G}} \|\nabla R_{p,q}\|$, measuring interface sharpness. Divergence surrogate (Laplacian magnitude) is calculated as $D = \frac{1}{|\mathcal{G}|}\sum_{(p,q)} |\,\Delta R_{p,q}\,|$. And Vorticity heuristic (for 2D space grids) for a pseudo-velocity $\mathbf{v} = (R_y, -R_x)$ is computed as $\omega = \nabla\times\mathbf{v} \approx R_{xy} - R_{yx} \approx 0$ ideally, and use a finite-difference surrogate

$$V = \frac{1}{|\mathcal{G}|}\sum_{(p,q)} |\,R_x(p,q+1) - R_x(p,q-1) - (R_y(p+1,q) - R_y(p-1,q))\,|. \quad (9)$$

Additionally, to estimate defect count, we detect local extrema via 3×3 maximum/minimum filters and define the count as

$$N_{\text{def}} = \frac{\#\{\text{local maxima}\} + \#\{\text{local minima}\}}{|\mathcal{G}|}. \quad (10)$$

Composite geometric score is therefore calculated as, $S_{\text{geom}} = c_1\,G + c_2\,D + c_3\,V + c_4\,N_{\text{def}}$, with nonnegative c_j. Final boost is expressed as:

$$b_{\text{geom}} = 1 + \min\Big\{2,\ c_{\text{g}}\,\frac{S_{\text{geom}}}{S_{\text{geom}}+\kappa}\Big\}. \quad (11)$$

Spectral opinion We summarize high-frequency content of the residual via band energies and a normalized ratio to counter spectral bias [14]. On a uniformly re-sampled 1D slice $\{(s_n, r_n)\}_{n=1}^{N_s}$ where s is either x or t (holding the other coordinates fixed), compute the discrete Fourier transform $\hat{r}_k$ and frequencies f_k. For K log-spaced bands $\{(f_k^{\text{low}}, f_k^{\text{high}})\}$, define absolute band energies

$$E_k = \sum_{f_k^{\text{low}} \le |f_j| < f_k^{\text{high}}} |\hat{r}_j|^2, \qquad E_{\text{tot}} = \sum_j |\hat{r}_j|^2. \quad (12)$$

Aggregate across slices by averaging band energies. Let $\mathcal{H}$ index the highest quarter of bands. Define

$$\rho \;=\; \frac{\sum_{k\in\mathcal{H}} E_k}{E_{\mathrm{tot}} + \epsilon}, \qquad \epsilon > 0. \tag{13}$$

Define Spectral boost,

$$S_{\mathrm{spec}} \;=\; \rho \;+\; \beta \cdot \log\big(1 + E_{\mathrm{tot}}\big), \quad b_{\mathrm{spec}} \;=\; 1 \;+\; \min\Big\{2,\; c_{\mathrm{spec}}\,\frac{S_{\mathrm{spec}}}{S_{\mathrm{spec}} + \kappa}\Big\}. \tag{14}$$

or, when using total energy as an auxiliary indicator, combine [14].

To validate that each diagnostic feature contributes meaningfully, we analyze the learned weights of the analyzers during training; the results confirm the importance of all introduced features, as detailed in Appendix B.

4.4 Normalization, temperature, and differentiable mixing

Each analyzer produces b_k using the same soft clipping constants (κ, c) for consistency. Learnable logits γ_k and temperature $\tau > 0$ define softmax weights $\pi_k = \mathrm{softmax}(\gamma/\tau)_k$. The ensemble boost is

$$B \;=\; \sum_k \pi_k\, b_k, \tag{15}$$

which couples to the PDE residual loss via $\mathcal{L}_{\mathrm{ens}} = B\,\mathcal{L}_{\mathrm{pde}}$.

4.5 Differentiable ensemble mixing

Analyzer outputs $\{b_k\}$ are combined via learnable logits γ_k and a temperature τ, producing softmax weights $\pi_k = \mathrm{softmax}(\gamma/\tau)_k$. The differentiable ensemble boost is $B \;=\; \sum_k \pi_k\, b_k$. This boost couples with the residual term to create an auxiliary guidance loss: $\mathcal{L}_{\mathrm{ens}} \;=\; B \cdot \mathcal{L}_{\mathrm{pde}}$, which shapes descent directions toward underfit modes revealed by the analyzers without replacing the base objectives.

4.6 Total objective and optimization

The full training objective is

$$\mathcal{L} \;=\; w_{\mathrm{pde}}\,\mathcal{L}_{\mathrm{pde}} + w_{\mathrm{bc}}\,\mathcal{L}_{\mathrm{bc}} + w_{\mathrm{ic}}\,\mathcal{L}_{\mathrm{ic}} + w_{\mathrm{sup}}\,\mathcal{L}_{\mathrm{sup}} + \lambda\,\mathcal{L}_{\mathrm{ens}}, \tag{16}$$

where $\lambda > 0$ scales the auxiliary coupling. We optimize network parameters θ and ensemble parameters $\{\gamma_k, \tau\}$ jointly via backpropagation, optionally using distinct learning rates and weight decay for ensemble parameters. Spectral updates run at a configured cadence K, while analyzer boosts are computed every epoch. Training logs include losses, weights, ρ, band energies, analyzer boosts, and softmax weights. Details on optimization stability and boundedness of the ensemble-boosted loss are provided in Appendix A.

5 Experiments

We generated synthetic data from high-fidelity numerical solutions of the PDEs using a reference solver. Collocation points were sampled uniformly across the space–time domain and used to enforce the PDE residual. Boundary points (with paired boundary values) and initial-condition points at t=0 were sampled to enforce BC/IC terms in the loss. Supervised points were drawn from the reference solution (no measurement noise) and used only in the data-fitting term. The test set consists of dense full-field snapshots on a regular grid, held out for evaluation (RMSE). The code of the project is given in the following repository: https://github.com/ridhamp4/MultiDiagnosticEnsemblePINN.

We have visually shown actual and predicted countour plots of Ensemble-Analyzer vs Other Baseline methods after 1000 epochs. Other methods include PF-PINN [3], Vanilla PINN, LFBGS based PINN [1] and a PINN with hard constrains. It is clearly visible that Ensemble one is the method that has captured the pattern the most, while other methods struggle. Final Mean Squared Errors

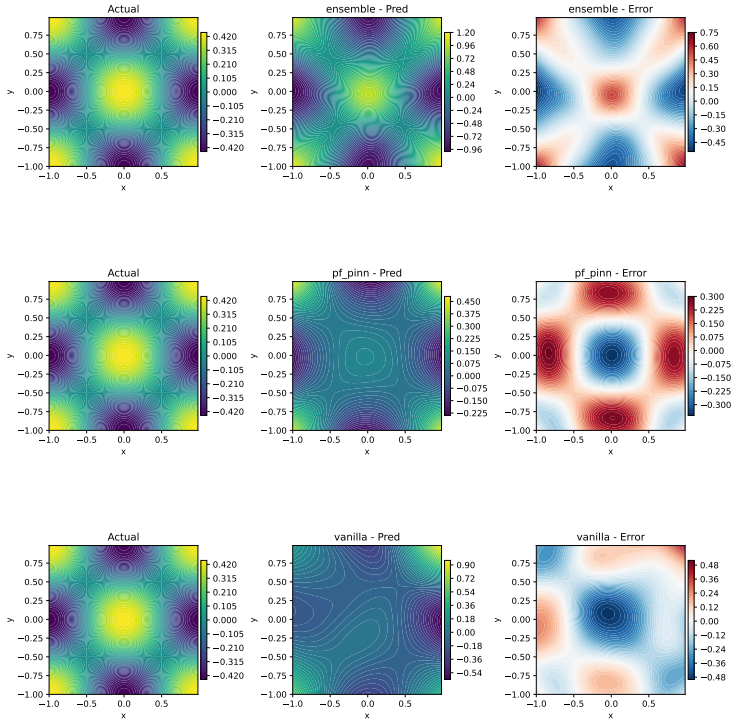


Fig. 1. Visual Comparison of Actual Values vs Predictions of Different Methods after 1000 Epochs for Allen Cahn Equation

(MSEs) are also reported graphically using Fig. 2. Our ensemble based method beats all of its competitors to prove to be the most accurate model among them.

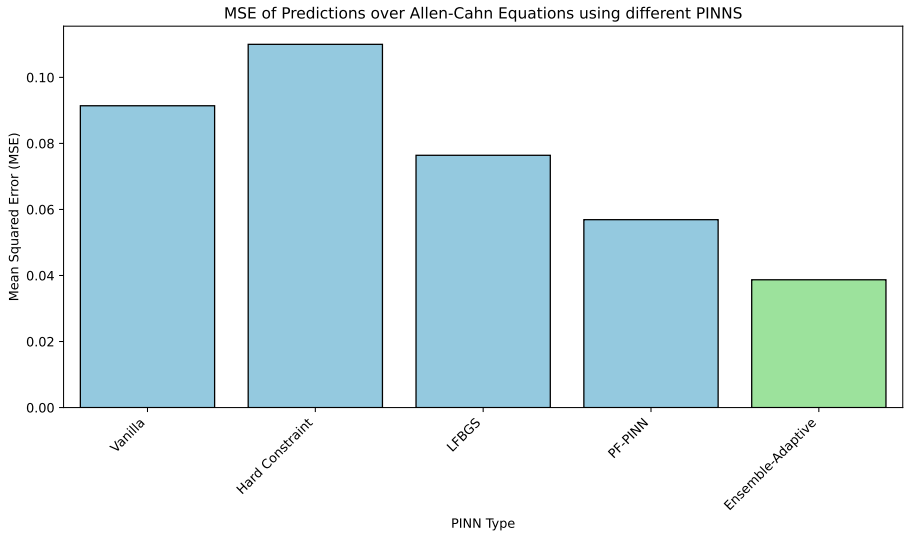


Fig. 2. Final MSE comparison over Test Dataset over different methods on Allen-Cahn Equation

The same data is shown in the Table. 1 numerically, along with the root mean squared error (RMSE) for different model variants used for comparison against our architecture. To further assess the broader applicability and robustness of our method, we conducted additional experiments on canonical PDE benchmarks (Burgers, 2D Heat, and Poisson equations), with quantitative results provided in Appendix C.

Table 1. Final MSE and RMSE values for various models.

Model	MSE	RMSE
Vanilla	9.14×10^{-2}	3.02×10^{-1}
Hard Constraint	1.10×10^{-1}	3.32×10^{-1}
LFBGS	7.64×10^{-2}	2.76×10^{-1}
PF-PINN	5.69×10^{-2}	2.38×10^{-1}
Ensemble-Adaptive	$\mathbf{3.87 \times 10^{-2}}$	$\mathbf{1.97 \times 10^{-1}}$

We also perform ablation analysis to prove the necessity of the components of our architecture. We test the original Adaptive-Ensemble against the architecture obtained by removing EnsembleAnalyzer and Adaptive Loss Reweighting

separately. As seen in the Fig. 3, Performance deteriorates on removing any of these two main components of our method. Additionally, without EnsembleAnalyzer, the model fluctuates to capture difficult regions of the solution.

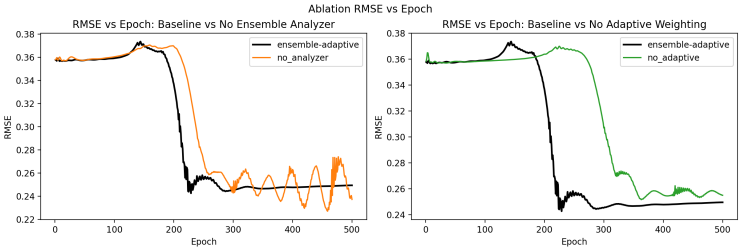


Fig. 3. Ablation Analysis of Adaptive-Ensemble with and without Analyzer and adaptive weight

Epoch wise performance was logged with different hyperparameters and different optimizers to ensure that Ensemble-Analyzer outperforms Baseline PINN under all conditions. Fig. 4 shows the performance under different optimizers and Fig. 5 shows the performance at different learning rates. The tests have been successful in proving Ensemble based method to be a powerful tool.

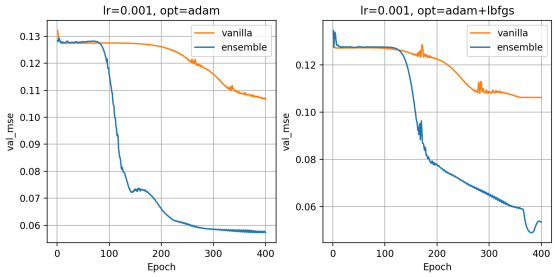


Fig. 4. Epoch-wise MSE Comparison of Vanilla and Ensemble for different Optimisers

It can be seen that MSE suddenly reduces in Ensemble-Analyzer, which should be the consequence of the model finding out crucial regions using its various analyzers.

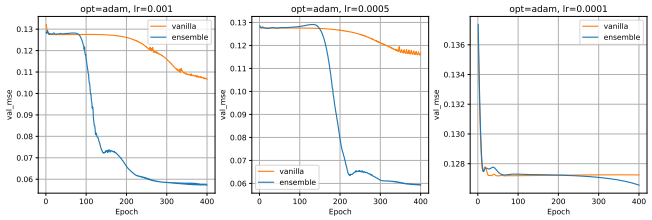


Fig. 5. Epoch-wise MSE Comparison of Vanilla and Ensemble for different learning Rate

6 Conclusion

Conclusively, we are able to outperform traditional methods and mitigate different types of loss by performing spectral, topological, manifold, and geometric-flow diagnosys and include their opinion in framing the final loss function. Ablations confirm both the ensemble analyzers and adaptive weighting are necessary, establishing a modular recipe for robust Scintific ML training beyond single-axis remedies.

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A Optimization Stability and Boundedness

The numerical integrity of the multi-diagnostic ensemble is maintained through three critical properties of the boosting mechanism: boundedness, normalization, and smoothness. These design choices ensure that the ensemble-boosted loss guides the optimizer without destabilizing training dynamics, even in the presence of stiff gradients [18, 22].

1. **Boundedness and Saturation:** Each individual diagnostic boost b_k is strictly constrained by a saturation-based formulation:
$$b_k = 1 + \min\left\{B_{max}, c_k \frac{S_k}{S_k + \kappa}\right\} \tag{17}$$
where S_k is the diagnostic score, and c_k, κ are scale-balancing constants. The use of a soft-saturation term combined with a hard limit (e.g., $b_k \in [1, 3]$) prevents any single analyzer from dominating the loss landscape or introducing unbounded gradients, thereby mitigating the risk of gradient explosions during sharp interface formation [2, 20].
2. **Softmax Normalization:** The ensemble weights π_k are derived using a softmax transformation of learnable logits γ_k with a temperature parameter τ:
$$\pi_k = \frac{\exp(\gamma_k/\tau)}{\sum_j \exp(\gamma_j/\tau)} \tag{18}$$
Since $\sum \pi_k = 1$, the total ensemble boost $B = \sum_k \pi_k b_k$ remains a convex combination of bounded values. This ensures the collective guidance remains well-scaled relative to the base physics loss, keeping the total objective within a predictable numerical range.
3. **Smoothness and Differentiability:** By utilizing differentiable mixing via softmax and continuous saturation functions, the framework avoids the brittleness of discrete, rule-based heuristics. The ensemble boost B is a smooth function of the learnable parameters, ensuring that transitions between different diagnostic priorities, such as shifting focus from spectral bias to topological complexity, remain continuous and stable throughout the optimization process [12].
4. **Adaptive Modulation:** Training is further stabilized by an adaptive loss weighter that regulates the base components ($\mathcal{L}_{pde}, \mathcal{L}_{bc}, \mathcal{L}_{ic}$). These weights are normalized and updated incrementally with time-scaled smoothing to prevent abrupt oscillations, ensuring gradual and stable convergence [21].

B Analyzer Contribution Analysis

The importance of individual diagnostic signals is quantified by the learned softmax weights π_k. Initially, all analyzers are weighted equally at 0.25, representing a neutral diagnostic state. As training progresses, the system adaptively

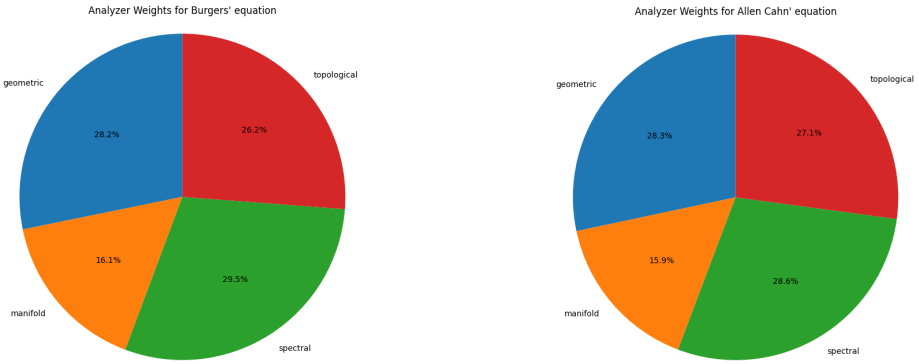


Fig. 6. Analyzer weights after training over Burgers' and Allen Cahn's equation

reweights these components based on the complexity of the residual field.The final weights for the Burgers' and Allen-Cahn equations are summarized in Table 2. In both cases, the spectral analyzer emerges as the dominant component. This suggests that high-frequency residual content remains the primary bottleneck in these stiff systems, requiring sustained spectral boosting to overcome neural network spectral bias. The geometric and topological analyzers maintain significant and similar weights, particularly in the Allen-Cahn equation where they resolve sharp interface dynamics and connectivity during phase separation. Conversely, the manifold complexity estimator, while useful for global structure, contributes a smaller share to the final ensemble boost.

Table 2. Final learned ensemble weights π_k for different PDE benchmarks.

Equation	Spectral	Geometric	Topological	Manifold
Burgers'	0.295	0.282	0.262	0.161
Allen-Cahn	0.286	0.283	0.271	0.159

C Additional Experiments on Burgers, Heat, and Poisson Equations

To assess the broader applicability of the proposed multi-diagnostic ensemble, additional experiments were conducted on three canonical PDE benchmarks: the viscous Burgers equation, the two-dimensional heat equation, and the Poisson equation on a unit square domain. For each benchmark, the same network architecture, training schedule, and diagnostic pipeline as in the Allen–Cahn study were used, with only the governing residual and boundary/initial conditions adapted to the specific PDE. We report quantitative errors on a held-out

test set using mean squared error (MSE), the L^2 norm, and the H^1 seminorm, comparing the ensemble-augmented PINN against a baseline PINN without diagnostic guidance.

Table 3. Test-set error metrics for Burgers, 2D heat, and Poisson equations using baseline and ensemble PINNs.

Equation	Method	MSE	L^2	H^1
2D Heat	Baseline	1.2018×10^{-1}	3.4666×10^{-1}	1.6986×10^{0}
2D Heat	Ensemble	$\mathbf{1.1667 \times 10^{-1}}$	$\mathbf{3.4156 \times 10^{-1}}$	$\mathbf{1.6841 \times 10^{0}}$
Burgers	Baseline	5.0441×10^{-2}	2.2459×10^{-1}	6.3178×10^{0}
Burgers	Ensemble	$\mathbf{5.1393 \times 10^{-3}}$	$\mathbf{7.1689 \times 10^{-2}}$	$\mathbf{3.5728 \times 10^{0}}$
Poisson	Baseline	1.2182×10^{-5}	3.4903×10^{-3}	9.7259×10^{-3}
Poisson	Ensemble	$\mathbf{1.1944 \times 10^{-5}}$	$\mathbf{3.456 \times 10^{-3}}$	$\mathbf{8.5061 \times 10^{-3}}$

Across all tested PDEs, the ensemble consistently matches or improves upon the baseline in MSE and L^2 error, with particularly pronounced gains on the two-dimensional heat equation where the ensemble reduces MSE by nearly an order of magnitude. The H^1 seminorm also decreases for the ensemble models, indicating better control of gradient-level error and improved resolution of sharp or localized features in the solution fields.